

YIJIA MA

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QUANTITATIVE PROFILE

Ph.D. statistician with extensive experience in stochastic processes, Bayesian inference, simulation, temporal point processes, and functional data alignment. Experienced in mathematical modeling, MCMC-based validation, uncertainty quantification, and Python/MATLAB/SQL workflows for noisy high-dimensional time-series data.

EDUCATION

Ph.D. in Statistics (GPA: 4.0/4.0) – Florida State University, FL	Aug 2024
- Dissertation: Bayesian modeling for functional data registration - Honors: Best First Year Student Award; Clara Kibler Davis Scholarship	
M.S. in Applied Mathematics and Statistics (GPA: 3.92/4.0) – Stony Brook University, NY	May 2017
B.S. in Actuarial Science and Risk Management – Tianjin University of Finance and Economics, China	Jun 2013

QUANT ANALYTICS CORE SKILLS

Modeling & statistical inference: Bayesian modeling; stochastic processes; Gaussian/non-Gaussian process priors; temporal point processes; functional data analysis; time-series modeling; statistical depth; rank-based methods

Simulation & risk analytics: Monte Carlo simulation; MCMC; resampling and permutation inference; uncertainty quantification; scenario and sensitivity analysis; model validation; stability and robustness assessment

Optimization & high-dimensional methods: Numerical optimization; gradient-based algorithms; penalized and regularized estimation; constrained optimization; high-dimensional feature selection; model tuning and calibration

Programming and data tools: Python (NumPy, Pandas, SciPy, scikit-learn, PyTorch, TensorFlow, NLTK), SQL, MATLAB, R, SAS, SPSS, Excel VBA, Tableau, AWS

PROFESSIONAL EXPERIENCE

Researcher / Data Scientist, University of Texas Health Science Center at Houston, TX	Aug 2024-Present
- Build end-to-end quantitative pipelines for high-dimensional neural and behavioral time-series data, covering data ingestion, preprocessing, feature extraction, statistical modeling, simulation-based validation, and reproducible reporting in Python and MATLAB. - Develop scalable signal-processing and model-validation workflows for noisy multivariate data, including normalization, automated artifact detection, ICA-based component separation, dynamic dependency analysis, and robust quality-control checks. - Design interpretable temporal, spectral, and connectivity features for downstream inference and supervised learning, translating complex time-series dynamics into decision-ready metrics for scientific stakeholders. - Train and evaluate regularized and nonlinear models, including L1/L2 logistic regression, SVM, random forests, and constrained neural-network architectures, emphasizing stable generalization under limited-sample and high-dimensional settings. - Apply resampling, permutation inference, mixed-effects modeling, and multiple-comparison control to distinguish reliable signal from noise and quantify model uncertainty. - Collaborate with investigators, engineers, and research staff to deliver accurate, documented, reusable analytical tools under evolving project requirements and fast timelines.	
Data Scientist Intern, Travelers, CT	Jun 2023-Aug 2023
- Developed a Natural Language Processing (NLP) analytics pipeline to identify overrepresented drivers of long customer-service calls, supporting operational efficiency and cost-optimization initiatives in an insurance business setting. - Built scalable AWS-based ingestion, cleaning, preprocessing, text-normalization, filtering, consistency-check, and anomaly-detection workflows for large-scale call transcript data. - Applied BERT-based question answering, sentence-transformer retrieval, GPT-based summarization, and topic modeling to classify call intents and improve interpretability for business users. - Performed statistical analyses of call length distributions, frequency imbalance, and overrepresentation patterns, converting unstructured data into actionable insights for stakeholders.	
Analyst, Vendor Team Services, NC	2017-May 2019
- Built Python-based web-scraping pipelines using Selenium and BeautifulSoup to collect large-scale customer-review data with 95%+ extraction accuracy. - Designed SQL-based ETL and reporting workflows for datasets exceeding 5 GB and automated recurring reporting with Excel VBA, improving reliability and reducing manual effort. - Delivered data-driven business recommendations by combining quantitative analysis, operational observations, and stakeholder communication.	

Stochastic Process Representation for Time Warping Functions

- Developed a linear Hilbert-space representation for nonlinear time-warping functions using centered log-ratio transformation, enabling Gaussian-process and broader stochastic-process modeling of phase variability under structural constraints.
- Demonstrated the representation through simulation studies, penalized function registration, gradient-based optimization, functional boxplots, and outlier detection in real temporal growth-pattern data.

Bayesian Function Registration and Uncertainty Quantification

- Built Bayesian registration frameworks that place stochastic-process priors directly on time warping functions, enabling Gaussian priors, efficient posterior inference
- Implemented MCMC / Z-mixture pCN simulation workflows to study prior effects, quantify alignment uncertainty, and evaluate robustness across simulated and real noisy functional data.
- Extended probabilistic registration models by defining likelihoods directly in the original function space, supporting robust signal estimation under compositional variation, additive noise, translation, and scaling effects.

Generative Modeling for High-Dimensional Time-Series Data

- Contributed to a flow-matching Transformer framework for high-fidelity EEG generation, modeling multi-channel time-series as continuous trajectories with long-range temporal and inter-channel dependencies.
- Incorporated structure-preserving constraints to maintain spectral structure, temporal stationarity, and signal-level statistics, improving distributional fidelity and downstream utility across large-scale benchmark datasets.

Customer Lifetime Value Prediction

- Cleaned and imputed 75K+ customer records, performed feature selection using filter, wrapper, and embedded strategies, and trained Random Forest, XGBoost, and LightGBM models with systematic hyperparameter tuning, achieving RMSE = 56.59.

SELECTED TOP JOURNAL PUBLICATIONS

- Ma, Y., & Wu, W. A New Framework for Bayesian Function Registration. *Electronic Journal of Statistics*, 2025.
- Wang, Y., Ma, Y., Li, W. & You, C. Let EEG Models Learn EEG. *Proceedings of the 43rd International Conference on Machine Learning*, 2026.
- Ma, Y., Zhou, X., & Wu, W. A Stochastic Process Framework for Time Warping Functions. *Computational Statistics and Data Analysis*, 2024.
- Ma, Y., & Wu, W. A Novel Point Process Model for Neuronal Spike Trains. *Frontiers in Applied Mathematics and Statistics*, 2024.
- Zhou, X., Ma, Y., & Wu, W. Statistical Depth for Point Process via the Isometric Log-Ratio Transformation. *Computational Statistics and Data Analysis*, 2023.
- Kim, W. M., Ma, Y., & Wu, W. Bayesian Registration of Functions with Compositional and Additive Noises. Preprint / under review